Gauri Trivedi

FPGA Design Engineer

RESUME_1816

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Total Experience: 6 years

PROFESSIONAL SUMMARY

FPGA Design Engineer with 6 years of experience in fpga, using VHDL, Verilog, Xilinx Vivado, Intel Quartus. Experienced in rtl simulation, fixed-point arithmetic and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: Hardware Simulation, AXI4, Verilog, DSP, Intel Quartus, VHDL, ILA, SPI, Tcl, FPGA, Xilinx Vivado, Python

Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

PROFESSIONAL EXPERIENCE

Senior FPGA Design Engineer | Asterline Engineering | Jan 2023 - Present

- Improved maximum operating frequency by 23% after timing-path optimization
- Verified 147 protocol corner cases with repeatable simulation suites
- Reduced FPGA resource utilization by 33% through arithmetic sharing
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Associate FPGA Design Engineer | OrionPath Technologies | Jul 2020 - Dec 2022

- Reduced FPGA resource utilization by 29% through arithmetic sharing
- Lowered end-to-end data-path latency by 14% using pipelined logic

PROJECTS

Real-Time Signal Filter | DSP, ILA, SPI, FPGA

- Implemented fixed-point FPGA pipelines with testbench golden models.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

High-Speed Data Logger | Tcl, Verilog, AXI4, Hardware Simulation

- Created DMA and buffering logic with ILA debugging hooks.
- Documented acceptance criteria and evaluated the design using fixed-point arithmetic and board bring-up.

EDUCATION

B.Tech in Electronics and Communication Engineering, Bayview Institute of Engineering, 2020
Relevant Methods: RTL simulation, Fixed-point arithmetic, Board bring-up

CERTIFICATIONS

Timing Closure Training (2021)